YEAR 10 PHYSICS, AQA GCSE, LESSON 2 OF 8

# PARTICLE MODEL OF MATTER

## LESSON 02

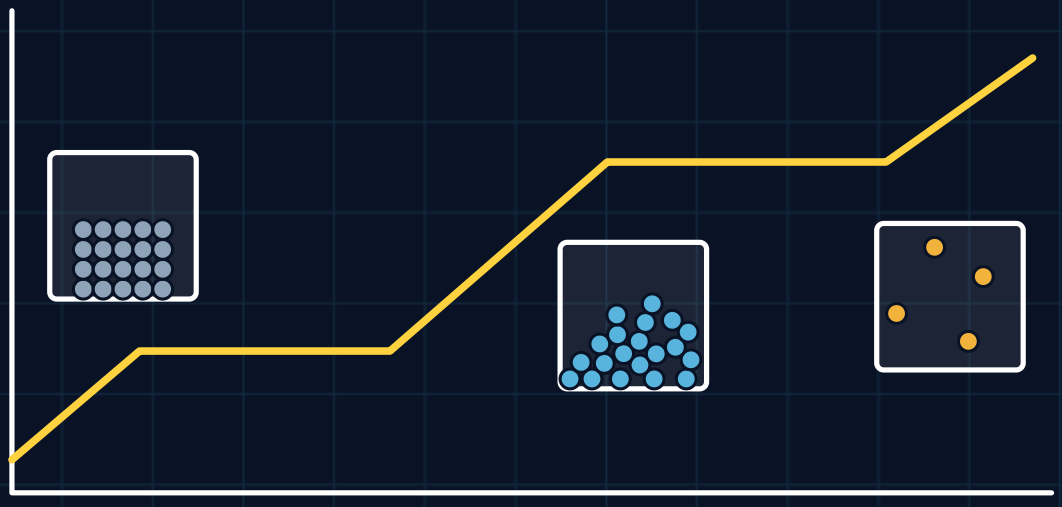
### GAS PARTICLES AND PRESSURE

An aerosol can says “Do not expose to temperatures above 50 °C”. What could go wrong?

**IN THIS LESSON YOU WILL**

- describe the motion of the molecules in a gas
- link the temperature of a gas to the average kinetic energy of its molecules
- explain how a gas pushes on the walls of its container
- explain why heating a gas at constant volume increases its pressure

**ANSWER EDITION FOR TEACHERS AND SELF-MARKING**



NAME _____ CLASS _____

START HERE

# HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 2** of the Particle Model of Matter unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

**DO NOW**
Quick questions from earlier lessons. Remembering again is what makes it stick.

**LEARN IT**
The explanation, diagrams and worked examples. Read this again when you revise.

**TRY IT**
Practice that builds up: first with support, then on your own, then a challenge.

**PRACTICAL**
Record your measurements and work out results from them.

**EXAM CORNER**
AQA-style questions with a tip on how the marks are given.

**REVISE IT**
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are close together **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

## THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

| | | | |
|---|---|---|---|
| 01 | Solids, liquids and gases | **02** | **Gas particles and pressure** Random motion, temperature and kinetic energy, what causes gas pressure, heating a gas at constant volume (slides 8–12) |
| 03 | Density | 04 | Measuring density (required practical) |
| 05 | Internal energy and heating curves | 06 | Specific latent heat |
| 07 | Specific heat capacity | 08 | Measuring specific heat capacity (required practical) |

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-model**

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Lesson

# 02 GAS PARTICLES AND PRESSURE

AQA Physics 4.3.3.1; Combined Science 6.3.3.1

**BIG QUESTION** **An aerosol can says “Do not expose to temperatures above 50 °C”. What could go wrong?**

**BY THE END YOU CAN**

- describe the motion of the molecules in a gas
- link the temperature of a gas to the average kinetic energy of its molecules
- explain how a gas pushes on the walls of its container
- explain why heating a gas at constant volume increases its pressure

**random motion** **kinetic energy** **temperature** **pressure** **collision** **constant volume**

**YOU WILL NEED**

Demonstration: a sealed round-bottomed flask joined to a pressure gauge, a large beaker or water bath, a kettle, a thermometer.

**Teaching deck, slides 8–12**
panphy.app/gcsephy/decks/particle-model/#8

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**
Describe the arrangement of the particles in a gas.

**Far apart, in a random arrangement.**

**2**
Why can a gas be squashed easily?

**There is a lot of empty space between its particles.**

**3**
Which energy store does a moving object have? Write the equation for it.

**The kinetic store.**

$E_k = \frac{1}{2} m v^2$

EXPLAIN

## A GAS NEVER KEEPS STILL

The molecules of a gas are in **constant random motion**. “Random” means that they move in all directions and at a range of speeds. They travel in straight lines until they hit each other or a wall, then bounce off.

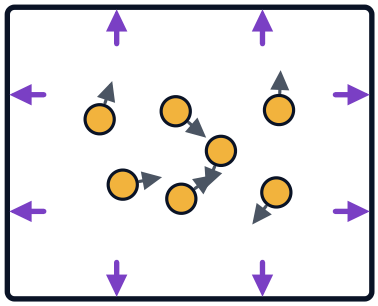
## TEMPERATURE AND KINETIC ENERGY

> ! The **temperature** of a gas is related to the **average kinetic energy** of its molecules. Heat a gas and its molecules move faster: the hotter the gas, the greater their average kinetic energy.

We say *average* because at any moment some molecules are moving fast and some slowly.

## WHAT CAUSES GAS PRESSURE?

1. The molecules **collide** with the walls of the container.
2. Each collision gives the wall a tiny **push** (a force).
3. There are billions of collisions every second, on every part of the wall.
4. Together they make a steady force on each square metre of wall: the **pressure** of the gas.

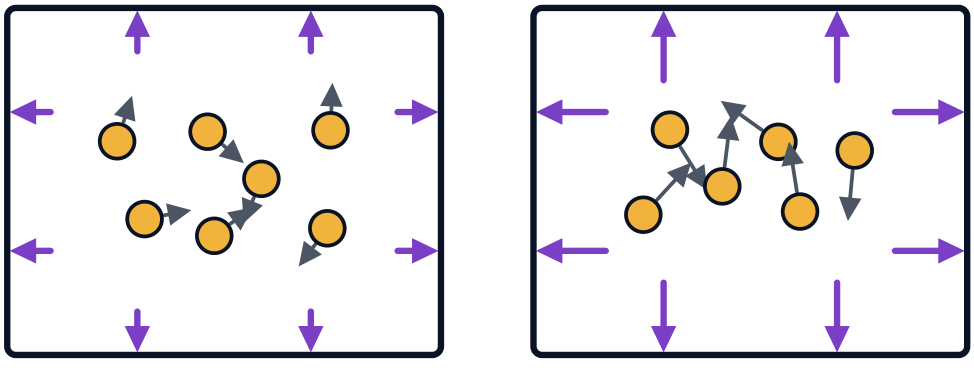


**gas in a sealed box**

→ push of the gas on the wall

## HEATING A GAS AT CONSTANT VOLUME

"Constant volume" means that the container is sealed and cannot get bigger, like a steel can. Follow the chain of reasoning:



**cooler gas** **hotter gas, same box**

→ push of the gas on the wall

1. The gas is heated, so the average kinetic energy of the molecules increases: they move **faster**.
2. They hit the walls **more often**.
3. Each collision pushes on the wall with a **bigger force**.
4. So the **pressure increases**.

> **Get the marks:** The molecules do **not** get bigger when a gas is heated, and they do not "expand". They move faster. Never write that particles expand.

Cooling does the opposite: the molecules slow down, hit the walls less often and less hard, and the pressure falls.

PRACTISE

## TRY IT

**1** Draw a ring around the correct answer in each box.

The molecules of a gas move

| in circles | **in random directions** | all in the same direction |
|---|---|---|

When a gas is heated, its molecules

| get bigger | **move faster** | move closer together |
|---|---|---|

The temperature of a gas is related to the molecules'

| **average kinetic energy** | size | mass |
|---|---|---|

**2** SUPPORTED Complete the sentences.

Gas pressure is caused by molecules **colliding with** (colliding with / sticking to / passing through) the walls of the container.

When a gas in a sealed can is cooled, the pressure **decreases** (increases / decreases / stays the same).

**3** A gas in a sealed steel can is heated. Put the statements in order, 1 to 4, to explain what happens.

| | |
|---|---|
| 3 | Each collision with the wall exerts a bigger force. |
| 1 | The average kinetic energy of the molecules increases, so they move faster. |
| 4 | The pressure of the gas increases. |
| 2 | The molecules hit the walls more often. |

**Order: 3, 1, 4, 2 (reading down the list).**

**4** A car is driven for a long time on a hot day. Explain why the pressure of the air in its tyres increases. Assume the volume of a tyre does not change.

**The air in the tyre gets hotter, so the molecules move faster (their average kinetic energy increases). They collide with the tyre wall more often and with more force, so the pressure increases.**

5 **CHALLENGE** A sealed bottle of air is put in a freezer. Mia says, “The pressure falls because some of the air particles disappear.” Explain why Mia is wrong, and give the correct reason.

**The bottle is sealed, so the number of particles stays the same. The pressure falls because the molecules move more slowly: they hit the walls less often and with less force.**

INVESTIGATE

## SEE IT: PRESSURE AND TEMPERATURE

Your teacher will demonstrate. The flask is sealed, so its volume stays the same.

A sealed flask of air is joined to a pressure gauge and put in water at different temperatures. Record the readings.

6 Record the results.

| Temperature of the air / °C | Pressure / kPa |
|---|---|
| 20 | 101 |
| 40 | 108 |
| 60 | 115 |
| 80 | 122 |

7 Describe the pattern in the results.

**As the temperature of the gas increases, its pressure increases.**

8 Which two things were kept the same in this demonstration?

**The volume of the gas (the flask is rigid) and the mass (or number of particles) of gas, because the flask is sealed.**

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

> **Get the marks:** A 3-mark “explain” about gas pressure needs three linked steps: **faster**, **collisions**, **pressure**.

**9** An aerosol can is left in sunlight and gets hot. Explain, in terms of particles, why the pressure of the gas in the can increases. **[3 marks]**

**The molecules gain kinetic energy and move faster (1). They collide with the walls of the can more often (1) and with a greater force, so the pressure increases (1).**

**10** Describe the motion of the molecules in a gas. **[2 marks]**

**They are in constant motion (1), in random directions (and with a range of speeds) (1).**

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- Gas molecules are in constant random motion.
- The temperature of a gas is related to the average kinetic energy of its molecules.
- Gas pressure is caused by the molecules colliding with the walls.
- Heating a gas at constant volume increases its pressure: faster molecules hit the walls more often and harder.

### COVER THE PAGE AND ANSWER

**1** What does “random motion” mean?

**2** What causes the pressure of a gas?

**3** A gas is cooled at constant volume. What happens to its pressure?

**4** What is the temperature of a gas related to?

### I CAN…

- describe the random motion of gas molecules
- link temperature to the average kinetic energy of the molecules
- explain how collisions with the walls cause pressure
- explain why heating a gas at constant volume increases its pressure

REFERENCE

# KEY WORDS FROM THIS LESSON

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

**Random motion**: Movement in all directions, at a range of speeds

**Gas pressure**: The force on each square metre of a surface, caused by gas molecules colliding with it

CHECK YOURSELF

# QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

### DO NOW

1. Far apart, in a random arrangement.
2. There is a lot of empty space between its particles.
3. The kinetic store. $E_k = \frac{1}{2}\,m\,v^2$

### COVER THE PAGE

1. The molecules move in all directions, at a range of speeds.
2. Its molecules colliding with the walls of the container.
3. It decreases.
4. The average kinetic energy of its molecules.